

DATA STRUCTURES AND ALGORITHMS

EEX4435

MINI PROJECT

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<u>Step 1 (Step one)</u>		
<i>Index i</i>	<i>ith word</i>	<i>Length of ith word</i>
1	Thehighoroororo	15
2	sample456ababcabab	18
3	ratesandhigh	12
4	qualityoroororooof	17
5	SDRsproproprocan	16
6	beusedintaaintto	16
7	exploretheethethe	17
8	relativetheethely	17
9	largerangeof	12
10	frequencies	11
11	ofthePJandRJ	12
12	inthepproprolight	19
13	sourceofthe	11
14	AMCWToFintaaintcameras	22
15	bysettinga	10
16	suitablelocal	13
17	oscillator	10
18	frequencytoget	14
19	alargeababcabab	15
20	intermediate	12

21	frequency(less	14
22	thantheNyquist	14
23	frequency)in	12
24	thedongleoroororo.	18
25	Also,this	9
26	conceptmaybe	12
27	usedtofindthe	13
28	jitterinthelight	16
29	sourceofthe	11
30	aforepropopro	14
31	mentioned	9
32	twomodulationtypes	18
33	ofToFrang	10
34	imagingtheethe	14
35	camerasaswell	13
36	asinanyother	12
37	relevantheethet	15
38	applications.	13
39	Inaddition,	11
40	itmaypossibleto	15
41	modifytheDSP	12
42	engineinoroororo	16
43	thedongle	9
44	whichisasoftware	16
45	basedcontroller	15

46	inordertointaint	17
47	getmaximum	10
48	benefitbut	10
49	thismaynot	10
50	beastraight	11
51	forwardababcabab	16
52	process.	8
53	In	2
54	Chapter7,	9
55	theinfluence	12
56	oftheperiodic	13
57	jittertheethe	13
58	onrangeintaint	15
59	measurementswas	15
60	Investigated.	13
61	Canweababcabab	14
62	compensate	10
63	theababcababPJ	14
64	amounttheethein	15
65	theoroororofirst	16
66	phasetep(by	12
67	determining	11
68	andcorrecting	13
69	it)before	9
70	itproceeds	10

71	to these second phase	16
72	step and soon,	12
73	in the correlation	16
74	function ab ab ab tion	17
75	within the	9
76	integration	11
77	period? With the	14
78	influence of the	14
79	PJ in each phase	13
80	step a constant (a	15
81	fixed the the	12
82	amount), then	12
83	it proper can be	16
84	further or or or	15
85	analysed	9
86	proper with the	16
87	cross talking	12
88	between phase	12
89	steps, during the	15
90	integration aintion	19
91	period aint. But	18
92	if it is not	9
93	a constant (	10
94	random behaviour	15
95	), then it	8

96	willbe	6
97	difficultto	11
98	compensate	10
99	fortheinfluenceof	17
100	thetheethePJonrange	19
101	measurements.	13
102	aretrying	9
103	proproprotodevelop	18
104	theAMCWToF	10
105	cameraswith	11
106	shortintegration	16
107	oroororoperiods.	16

STEP 2 (Step two)

a) State briefly what is the Boyer-Moore algorithm that uses in pattern matching operations

The Boyer-Moore algorithm is a fast pattern matching algorithm used to find a substring(pattern) inside a larger string (text). The Boyer-Moore algorithm matches a pattern against a text by comparing characters from right to left and uses two clever heuristics-the Bad character rule and Good Suffix Rule- to skip sections of the text , making it much faster than simple brute-force matching.

b) Describe briefly, what are the two approaches, namely Bad Character Heuristic and Good Suffix Heuristic.

1.Bad Character Heuristic-

When a mismatch occurs during comparison(from right to left) the algorithm checks the mismatched character in the text. It then shifts the patterns so that:

- The rightmost occurrence of that mismatched character in the pattern aligns with it, or
- If the character does not exist in the pattern, the pattern is shifted past that character.

2. Good Suffix Heuristic-

When a mismatch occurs but some part at the end of the pattern matched (called a “good suffix”), the algorithm shifts the pattern so that:

- Another occurrence of that suffix in the pattern aligns with the matched text, or
- If the suffix does not appear elsewhere, the pattern shifts to align the longest prefix of the pattern with the text.

<i>Index ,j</i>	<i>jth</i> <i>pattern</i>	<i>BCT for jth pattern</i>
1	anan	a:1 n:1 *:4
2	ith	i:2 t:1 h:1*:3
3	ababcabab	a:1 b:1 c:4 *:9
4	as	a:1 s:1 *:2
5	on	o:1 n:1 *:2
6	th	t:1 h:1 *:2
7	oroororo	o:1 r:1 *:8
8	tt	t:1 *:2
9	to	t:1 o:1 *:2
10	and	a:2 n:1 d:1 *:3
11	ter	t:2 e:1 r:1 *:3
12	propropro	p:2 r:1 o:1 *:9
13	theethe	t:2 h:1 e:1 *:7
14	ent	e:2 n:1 t:1 *:3
15	tio	t:2 i:1 o:1 *:3
16	mera	m:3 e:2 r:1 a:1 *:4
17	tter	t:2 e:1 r:1 *:4
18	freq	f:3 r:2 e:1 q:1 *:4
19	the	t:2 h:1 e:1 *:3
20	intaaaint	i:2 n:1 t:1 a:3 *:8

Using

Max(1 , length of pattern-index of pattern - 1)

STEP 3(Step three)

<i>Patt ern inde x,j</i>	<i>Jth patt ern</i>	<i>Words</i>	<i>Index range</i>
1	anan	No matching words	
2	ith	withinthe period?Withthe proproprowiththe cameraswith	2-4 9-11 11-13 9-11
3	ababc abab	sample456ababcabab alargeababcabab forwardababcabab Canweababcabab theababcababPJ funcababcababtion	10-18 7-15 8-16 6-14 4-12 5-13
4	as	AMCWToFintaaintcameras camerasaswell asinanyother whichisasoftware basedcontroller beastraight measurementswas	21-22 6-7 and 8-9 1-2 8-9 2-3 3-4 3-4 and 14-15

		phasetest(by	3-4
		tothesecondphase	14-15
		PJineachphase	11-12
		analysed	6-7
		betweenphase	10-11
		measurements.	3-4
		cameraswith	6-7
5	on	thedongleororo.	5-6
		conceptmaybe	2-3
		mentioned	6-7
		twomodulationtypes	12-13
		applications.	10-11
		Inaddition,	9-10
		thedongle	5-6
		basedcontroller	7-8
		onrangeintaaint	1-2
		tothesecondphase	9-10
		stepandsoon,	10-11
		inthecorrelation	15-16
		funcababababtion	16-17
		integration	10-11

		step a constant(a	7-8
		integrating a function	18-19
		a constant(	3-4
		the the the P on range	13-14
		short integration	15-16
6	th	explore the the the	8-9 and 12-13 and 15-16
		relative the the the	9-10 and 13-14
		of the P and R J	3-4
		in the pro pro pro light	3-4
		source of the	9-10
		than the Nyquist	1-2 and 5-6
		the dangle or or or or.	1-2
		Also, this	6-7
		used to find the	11-12
		jitter in the light	9-10
		source of the	9-10
		imaging the the the	8-9 and 12-13
		as in any other	9-10
		relevant the the the	8-9 and 12-13
		modify the DSP	7-8
		the dangle	1-2

	this may not	1-2
	the influence	1-2
	of the periodic	3-4
	jitter the the	7-8 and 11-12
	the abab cabab PJ	1-2
	amount the the in	7-8 and 11-12
	the or or or first	1-2
	to the second phase	3-4
	in the correlation	3-4
	within the	3-4 and 7-8
	period? With the	10-11 and 12-13
	influence of the	12-13
	fixed the the	6-7 and 10-11
	amount), then	9-10
	further or or or or	4-5
	pro pro pro with the	12-13, and 14-15
	steps, during the	13-14
	), then it	3-4
	for the influence of	4-5
	the the the PJ on range	1-2 and 4-5 and 8-9

		theAMCWToF	1-2
		cameraswith	10-11
7	oro oro	Thehighorooro	8-15
		qualityorooroof	8-15
		thedongleorooro	10-17
		engineinorooro	9-16
		theorooroofirst	4-11
		furtherorooro	8-15
		oroorooperiods.	1-8
8	tt	beusedintaaintto	14-15
		bysettinga	5-6
		jitterinthelight	3-4
		jittertheethe	3-4
		amounttheethein	6-7
		difficultto	9-10
9	to	beusedintaaintto	15-16
		oscillator	8-9
		frequencytoget	10-11
		usedtofindthe	5-6
		itmaypossibleto	14-15
		inordertoaint	8-9

		to these second phase	1-2
		difficult to	10-11
		proper proto develop	10-11
10	and	rates and high	6-8
		of the PJ and RJ	8-10
		and correcting	1-3
		step and soon,	5-7
		random behaviour	2-4
11	ter	intermediate	3-5
		jitter in the light	4-6
		Chapter 7,	5-7
		jitter the the	4-6
		determining	3-5
12	proper opro	SDRs proper pro can	5-13
		in the proper pro light	6-14
		afore proper pro	6-14
		it proper pro can be	3-11
		proper pro with the	1-9
		proper proto develop	1-9
13	the the he	explore the the the	8-14
		relative the the the	9-15
		imaging the the the	8-14

		relevant the the	8-14
		jitter the the	7-13
		amount the the in	7-13
		fixed the the	6-12
		the the the P on range	4-10
14	ent	mentioned	2-4
		measurements was	9-11
		measurements.	9-11
15	tio	mentioned	4-6
		two modulation types	10-12
		applications.	8-10
		In addition,	7-9
		in the correlation	13-15
		funcabab cabab tion	14-16
		integration	8-10
		integratinta intion	16-18
		short integration	13-15
16	mera	AMCW To Fintaaint cameras	18-21
		cameras as well	3-6
		cameras with	3-6
17	tter	jitter in the light	3-6
		jitter the the	3-6

18	freq	frequencies	1-4
		frequencytoget	1-4
		frequency(less	1-4
		frequency)in	1-4
19	the	exploretheethe	8-10 and 12-14 and 15-17
		relativetheethely	9-11 and 13-15
		ofthePJandRJ	3-5
		inthepproprolight	3-5
		sourceofthe	9-11
		thantheNyquist	5-7
		thedongleoroororo.	1-3
		usedtofindthe	11-13
		jitterinthelight	9-11
		sourceofthe	9-11
		imagingtheethe	8-10 and 12-14
		asinanyother	9-11
		relevantheethet	8-10 and 12-14
		modifytheDSP	7-9
		thedongle	1-3
		theinfluence	1-3

		oftheperiodic	3-5
		jittertheethe	7-9 and 11-13
		theababcababPJ	1-3
		amounttheethein	7-9 and 11-13
		theoroororofirst	1-3
		tothesecondphase	3-5
		inthecorrelation	3-5
		withinthe	7-9
		period?Withthe	12-14
		influenceofthe	12-14
		fixedtheethe	6-8 and 10-12
		amount),then	9-11
		furtheroroororo	4-6
		proproprowiththe	14-16
		steps,duringthe	13-15
		),thenit	3-5
		fortheinfluenceof	4-6
		thetheethePJonrange	1-3 and 4-6 and 8-10
		theAMCWToF	1-3
20	intaaint	beusedintaaintto	7-14

		AMCWToFintaaintcameras	8-15
		inordertointaaint	10-17
		onrangeintaaint	8-15
		integratintaaintion	9-16
		periodintaaint.But	7-14

(b) Compute the time complexities for the best-case and worst-case for your BM algorithm.

- Best-Case Time Complexity: $O(n/m)$ or $O(n/m + m) \approx O(n/m)$

In the best case, the pattern characters do not appear frequently in the text, allowing large jumps due to the bad character or good suffix shifts

For an example when the first mismatched character appears early in each comparison, the pattern can shift nearly its full length on each move.

Example-

Searching pattern "ABCD" in text "ZZZZZ"

Each mismatch allows a shift of $m(=4)$ characters, so only about n/m comparisons are needed.

The algorithm runs sub-linearly. It does not examine every character of the text.

- Worst case time complexity: $O(n \times m)$

Occurs when the pattern and text share many overlapping prefixes or repetitive substrings. (ex-"GGG" in "GGGGG").

In real text data, Boyer–Moore performs much better than this bound and is regarded as one of the **fastest string-search algorithms** in practice.

(c) Explain what is meant by the shifting rules in the BM algorithm.

"Shifting rules " determines how far the pattern moves (shift) to the right after a mismatch is detected during the comparison.

The two shifting heuristics-

1. Bad character rule

- When a mismatch occurs at the character $T[i]$ in the text and $P[j]$ in the pattern, the pattern is shifted so that the last occurrence of $T[i]$ in the pattern aligns with that position.
- If $T[i]$ does not appear in the pattern , the pattern can shift past the mismatched character entirely.
- Reduces unnecessary comparisons by skipping over irrelevant text characters.

2. Good suffix rule-

- If a suffix of the pattern has matched successfully before a mismatch occurs, shift the pattern so that the next possible occurrence of that suffix(or part of it) in the pattern aligns with the text.

- If that suffix doesn't appear elsewhere in the pattern, shift so that the next character after the suffix aligns.
- This makes additional jumps when the same substring repeats within the pattern.

Boyre-Moore uses both heuristics together (the larger of the two shifts) to maximize the jump distance and minimize redundant checks.

(d) State what other algorithms can be used for pattern-matching operations.

1. Naive string matching-

- Compares the pattern at every position in the text sequentially.
- Time complexity- $O(n \times m)$
- Simple, easy to implement for small data

2. Knuth-Morris-Pratt(KMP)-

- Uses a prefix function(LPS array) to avoid rechecking characters that are already matched
- Time complexity- $O(n \times m)$
- Useful when many patterns share prefixes

3. Rabin-Karp algorithm

- Uses hashing to compare pattern and text substring efficiently.
- Time complexity- Average $O(n + m)$, Worst $O(n \times m)$
- Good for multiple pattern search

4. Aho-Corasick algorithm

- Builds a trie(finite-state machine) for multiple patterns
- Time complexity- $O(n + z)$ (where z = no. of matches)
- Ideal for searching many patterns simultaneously.

5. Finite-Automation matcher

- Precomputes transitions for every character (state machine)
- Time complexity- $O(n + m \times \sigma)$
- Fast in fixed alphabets (ex. DNA sequences)